

LECTURE SCRIBE – CSE400 (Lecture 18)

Topics

- Marginal PMF
- Correlation
- Covariance
- Pearson's Coefficient
- Joint Gaussian Random Variables

1 Marginal PMF

Definition

Key Idea (from slide):

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable.**

Marginal PMF from Joint PMF Table

Step-by-step reasoning:

Let joint PMF be $P(X = x, Y = y)$

To find marginal of X :

1. Fix a value of x
2. Sum over all possible values of y

$$P_X(x) = \sum_y P(X = x, Y = y)$$

To find marginal of Y :

1. Fix a value of y
2. Sum over all possible values of x

$$P_Y(y) = \sum_x P(X = x, Y = y)$$

Example (as given in slides)

Given:

Joint PMF

Solution steps (as shown):

1. Identify all values of joint PMF
2. For each x , sum across all y
3. For each y , sum across all x

Final conclusion:

Marginal PMF is obtained by **systematically summing joint probabilities along rows or columns**.

2 Correlation (Raw Correlation)

Definition

A basic way to measure joint behavior is through the **product moment**:

$$R_{XY} = E[XY]$$

Interpretation

- Represents the **average value of the product** (XY)
- Called a **raw correlation measure**

Important Observations (from slides)

- Depends on **scale of X and Y**
- Computed **about the origin** (not about means)
- Mixes:
 - Mean values
 - Spread
 - Dependence

Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

Transition Idea

To remove the effect of means:

- **Center the variables**
- Leads to **Covariance**

3 Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step-by-Step Interpretation

1. Subtract mean from each variable:

- $(X - E[X])$
- $(Y - E[Y])$

2. Multiply deviations:

$$(X - E[X])(Y - E[Y])$$

3. Take expectation:

$$E[(X - E[X])(Y - E[Y])]$$

Meaning (from slides)

- Measures how **centered quantities vary together**
- Captures whether deviations occur:
 - In same direction
 - In opposite direction

4 Algebraic Form of Covariance

Start with definition:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Expand step-by-step:

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

Apply expectation:

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

Simplify:

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

5 Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

3. Linearity with Respect to Constants

Covariance is linear when constants are involved.

4. Independence Property

If X and Y are independent:

$$\text{Cov}(X, Y) = 0$$

Important conclusion (from slide):

- Zero covariance $\Rightarrow$ **uncorrelated**
- Does **not** imply independence

6 Covariance Example 1

Given:

Expressions for variables

Solution Steps (exact slide flow):

1. Use covariance formula:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

2. Substitute given expressions into formula
3. Expand expression
4. Compute expectations
5. Substitute values into covariance formula
6. Simplify

Final Result:

Obtained after substituting and simplifying (as shown in slides).

7 Pearson's Correlation Coefficient

Definition

- Measures **effect of change in one variable when the other changes**

Key Properties

- Measures:
 - Strength
 - Direction of linear relationship
- It is:
 - **Normalized covariance**
 - **Unit-free**

8 Covariance Example 2 (Continuous Case)

Given:

Joint PDF $f(x, y)$

Required:

Compute required quantities (expectations, covariance)

Solution Flow (as per slides)

1. Start from joint PDF
2. Compute marginal densities
3. Compute expectations:

$$E[X], \quad E[Y]$$

4. Compute:

$$E[XY] = \int \int xyf(x, y) dx, dy$$

5. Substitute into:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

6. Evaluate integrals step-by-step

Final:

Covariance obtained after full substitution and simplification (as shown across slides).

9 Joint Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if:

- Joint density has **bivariate Gaussian form**
- Marginals are **Gaussian**

Interpretation

- Used to model variables affected by many random factors

Real-life Example (from slide)

- (X) : CO₂ concentration
- (Y) : Temperature

Their joint behavior can be approximated by a **joint Gaussian distribution**.

Applications

- Signal processing
- Communication systems
- Noisy measurements

Key Idea

Joint Gaussian RVs:

- Represent **continuous random behavior**
- Capture dependence through covariance

Final Revision Flow

1. Start with **Joint PMF** $\rightarrow$ **Marginal PMF** (summation)
2. Measure joint behavior using:
 - $E[XY]$ (raw correlation)
3. Improve measure by centering:
 - Covariance
4. Normalize:
 - Pearson coefficient
5. Extend to continuous case:
 - Integrals
6. Model real systems:
 - Joint Gaussian RVs